

Week 9 Discussion: Binomial Trees (No-Arbitrage Pricing)

Main focus (50min): (i) one-step replication & risk-neutral pricing; (ii) two-step pricing via backward induction; (iii) American options and early exercise; (iv) CRR tree construction (matching volatility).

Module A) One-Step Binomial Model: Replication & Risk-Neutral Pricing

Model: $S_0 \rightarrow \begin{cases} S_0 u \\ S_0 d \end{cases}$ over T years, with $u > 1$, $d < 1$, risk-free rate r (cont. comp.).

No-arbitrage condition: $d < e^{rT} < u$.

Option payoffs: $f^u = h(S_0 u)$, $f^d = h(S_0 d)$.

Replication (static over one step):

$$\Delta_0 = \frac{f^u - f^d}{S_0 u - S_0 d}, \quad f_0 = \Delta_0 S_0 - B_0,$$

where B_0 is the amount *borrowed* (if $B_0 > 0$) or *invested* (if $B_0 < 0$) in the bank.

Risk-neutral probability and pricing:

$$p = \frac{e^{rT} - d}{u - d}, \quad f_0 = e^{-rT} (p f^u + (1 - p) f^d).$$

Interpretation: under p , the expected growth of the stock is the risk-free rate: $\mathbb{E}^Q[S_T] = S_0 e^{rT}$.

Practice A (Textbook 13.4): One-step call pricing via no-arbitrage

A stock price is currently $S_0 = \$50$. It is known that at the end of 2 months it will be either \$53 or \$48. The risk-free rate is 10% per annum with continuous compounding. What is the value of a 2-month European call option with strike $K = \$49$? Also report the hedge ratio Δ_0 .

Quick Tree Construction (CRR, constant parameters)

When T is split into N equal steps of size $\Delta t = T/N$ (in years), a common choice is

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}, \quad p = \frac{e^{r\Delta t} - d}{u - d}.$$

Then price by backward induction with $f = e^{-r\Delta t} (p f^u + (1 - p) f^d)$ at each node.

Module B) Two-Step Trees: Backward Induction & Dynamic Delta

Two steps: $t = 0, \Delta t, 2\Delta t$. At any node,

$$f = e^{-r\Delta t}(pf^u + (1-p)f^d).$$

Dynamic hedging: the hedge ratio must be *rebalanced* after the first move. At a node with next-step stock prices S^u, S^d and option values f^u, f^d ,

$$\Delta = \frac{f^u - f^d}{S^u - S^d}.$$

Concept check (Textbook 13.3): Why can you *not* choose a *single* (Δ, cash) at $t = 0$ so that the position stays riskless for the whole life of a two-step option?

Practice B (Textbook 13.7): Two-step European call

A stock price is currently $S_0 = \$50$. Over each of the next two 3-month periods it is expected to go up by 6% or down by 5%. The risk-free rate is 5% per annum with continuous compounding. What is the value of a 6-month European call option with strike $K = \$51$?

Module C) American Options: Early Exercise Test (Backward DP)

At maturity: $f_T = \text{payoff}$.

At each earlier node:

$$\text{continuation} = e^{-r\Delta t}(pf^u + (1-p)f^d), \quad \text{intrinsic} = \text{payoff if exercised now.}$$

$$f = \max\{\text{intrinsic}, \text{continuation}\}$$

Early exercise is optimal at a node only if $\text{intrinsic} > \text{continuation}$.

Practice C (Textbook 13.8): European put + parity check + American early exercise

For the situation considered in **Practice B (13.7)**, what is the value of a 6-month **European put** option with a strike price of $K = \$51$? Verify that the European call and European put prices satisfy put-call parity. If the put option were **American**, would it ever be optimal to exercise it early at any of the nodes on the tree?